Natural Language Processing

Lecture notes from my graduate level Quantum Computing class taught by professor James H. Martin at the University of Colorado Boulder. This course covers a range of topics in natural language processing including, but not limited to: probabilistic language modeling, text classification, vector semantics, neural networks, large language models, and transformer architecture. The textbook used is the 3rd Edition of *Speech and Language Processing* by Dan Jurafsky and James H. Martin. The draft of the 3rd edition can be found here: `https://web.stanford.edu/~jurafsky/slp3/`

Disclaimer: This document will contain mistakes such as typos, errors, etc. These were taken during the process of learning a new subject, so do what you will with that information.

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Contents

1 What is NLP?

- any language tech involved with generating, manipulating, etc.
  - e.g. spellcheck, dictation, translation, text-to-speech
- **Computational Linguistics**: modeling human language using computers so we can understand how languages work
  - linked to psychology, philosophy, and cognitive science.

1.1 The Three Pillars of Modern NLP

- **Probabilistic language models**: systems that assign probability distributions over possible next words.
- **Neural Networks**: the fundamental computational mechanism. Modeled after neurons turning into deep architectures optimized via gradient descent.
- **Embeddings:** vector-based representations where meaning is mapped as a point in high-dimensional continuous space.

2 Words, Lexicons, Morphology

- drives LLMS these days
- fundamental bridge from linguistic forms to underlying conceptual representations
- **type**: entries in vocab/lexicon
- how do we manage words/tokens?

2.1 Morphology

- the study of the ways words are built up from smaller units called morphemes.
- **Two classes of morphemes**:
  - **stems**: the core meaning-bearing unit
  - **affixes**: bits and pieces that attach to stems but do not change the core meaning of the word example: "star_s" with the core being "star" and the "-s" being the plural affix we attach.

2.1.1 Inflectional morphology

the combination of stems and affixes where the resulting word...

- has the same word class
- serves a grammatical purpose that is
  - different from the original
  - transparently related to the original
    - "walk" + "ing" = walking
    - "cat" + "s" = cats

2.1.2 Derivational morphology

messy stuff that nobody ever "taught" when learning english

- wide variety of affixes
  - -al, -tion, -ee, -er, -ant, -ness
  - anti-, un-, non-
- quasi-systematic rules
- irregular meaning change
- changes of word class

2.2 Language classification

- **Genetically**: groups based on family history
- **Areal**: geographical occurences
- **Typological**: shares grammar, morphology, phonology, syntax, etc.

2.3 Morphological Typology

groups languages based on aspects of morphology

Isolating vs Synthetic

- **Isolating**: one morpheme per word
- **Synthetic**: lots of morphemes per word

Agglutinating vs fusional

- **Agglutinating**: form words by concatenating stems with affixes to form words (with prefixes, suffixes, infixes)
- **Fusional**: allow morphemes to change form upon inflection; harder to isolate them

2.3.1 Computational morphology

Morphological analysis takes surface form and returns the stem with a set of morphological features rather than segmenting the surface form.

2.4 Vocabulary

- a vocabulary is the set of words a system has access to
- the minimal units of processing in an input that has some semblance of meaning attached to it.

What should such a vocabulary (or lexicon) consist of?

- all the words in a language?
- all the words in some comprehensive dictionary?
- some subset?

2.5 Corpora

1. Collect a big bunch of text.
2. Figure out the words that appear in it.
3. That's your vocabulary

Dimensions

- **Language**: dialects, creoles, pidgins, etc.
- **Provenance**: how was it obtained? Consent? Legal?
- **Genre**: Newswire, fiction, non-fiction, scientific articles, Wikipedia
- **Demographics**: Writer's age, gender, race, socioeconomic status, etc.
- **Medium**: Spoken, written, captioned, video
- **Writing system**: Over 350 systems in use

How many words are there? N = number of mentions V = vocabulary = set types ($|V|$ is the size of the vocab)

2.6 Heap's Law

$$|V| = kN^{\beta}$$

Where k and β are free variables. Usually k between 10 and 100 and β between 0.6 and 0.7.

- Size of the vocabulary grows <u>somewhat faster</u> that the square root of the corpus size
- The rate of the vocabulary growth slows as corpus grows, but never <u>flattens out</u>.

2.7 Issues in Normalization/Tokenization

- Hong Kong → one word? two?
- m.p.g → is this even a word?
- IRA, I.R.A → expand? keep out?

2.8 Sentence Segmentation

Tokenization, morphology, and further syntactic processing first requires sentence segmentation

- Easy in English. Punctuation is used to mark sentence boundaries.
  - "!", "?" are relatively unambiguous.
  - Periods can be. Like "Dr.", "I.R.A", etc.

2.9 Multilingual issues in Normalization

English is not the only language out there, and there are several problems that need to be faced.

- Code switching – "Chinglish", "Spanglish", etc.
- Writing systems – Latin, Chinese, Arabic
- Complex morphology –
- Sentence segmentation – Chinese, Thai, Japanese do not use spaces to separate words